



MECH MAINT- TURBINE Department
SIMHADRI

Note No :1



Ref: SMTTP/MECH MAINT-/2023-24/SIM/TMD/455280

Date:

05/09/2023(12:52:14)

Subject: Approval of LMI on Inspection & Maintenance of Condenser on load tube cleaning system : reg

Subject: Approval LMI on Inspection & Maintenance of Condenser on load tube cleaning system_Reg:

LMI: "Inspection & Maintenance of Condenser on load tube cleaning system (COLTCS)", with LMI NO. LMI/OGN/OPS/MECH/054 based on the OGN by CC-OS is due for revision.

No changes are envisaged.

Enclosed:

1. LMI "LMI-INSPECTION AND MAINTENANCE OF COLTCS REV00"
2. OGN "COS-ISO-00-OGN-OPS-MECH-054"

Put up for approval of competent Authority.

Submitted by:

रश्मि रंजन सतपथी - वरिष्ठ प्रबंधक

RASHMI RANJAN SATAPATHY - 100837 (SENIOR MANAGER ,MECH MAINT- TURBINE)

,Simhadri

(RRSATPATHY@NTPC.CO.IN, 9437021584)

On:05/09/2023(12:52:14) **Ref:**SMTTPP/MECH MAINT-/2023-24/SIM/TMD/455280

Note No:2

Forwarded for approval.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,OPERATION)

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On:05/09/2023(15:13:28) **Ref:**SMTTPP/MECH MAINT-/2023-24/SIM/TMD/455280

Note No:3

Forwarded

Submitted by:

शंभू कुमार सुमन - अपर महाप्रबंधक

Shambhu Kumar Suman - 005007 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE)

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On:05/09/2023(16:23:40) **Ref:**SMTTPP/MECH MAINT-/2023-24/SIM/TMD/455280

Note No:4

Operation, C&I and EMD HODs may please go through the revised LMI and comment (if any) before the same is got approved.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoov - 005637 (GENERAL MANAGER ,MAINTENANCE)
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On:13/09/2023(11:10:35) **Ref:**SMTTPP/MECH MAINT-/2023-24/SIM/TMD/455280

Note No:5

For review of LMI especially w.r.to the roles and responsibilities defined for Operation dept in the LMI

Submitted by:

टी.आर. राजगोपालन् - महाप्रबंधक

TR Rajagopalan - 003994 (GENERAL MANAGER ,OPERATION)
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On:13/09/2023(11:26:25) **Ref:**SMTTPP/MECH MAINT-/2023-24/SIM/TMD/455280

Note No:6

Proposed LMI is reviewed and found in line with OGN.

Submitted by:

मलय कांति जाना - अपर महाप्रबंधक

Malay Kanti Jana - 007824 (ADDL. GENERAL MANAGER ,OPERATION)
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On:18/09/2023(14:10:57) **Ref:**SMTTPP/MECH MAINT-/2023-24/SIM/TMD/455280

Note No:7

LMI on Inspection and Maintenance of Condenser on load tube cleaning system (COLTCS) reviewed and put up for approval.

Submitted by:

जी एच राव - अपर महाप्रबंधक

G H Rao - 006589 (ADDL. GENERAL MANAGER ,C & I MAINT- BOILER)
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On:18/09/2023(15:53:36) **Ref:**SMTTPP/MECH MAINT-/2023-24/SIM/TMD/455280

Note No:8

Review

Submitted by:

धर्मेन्द्र कुमार तोमर - अपर महाप्रबंधक

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On:18/09/2023(18:13:28) **Ref:**SMTTPP/MECH MAINT-/2023-24/SIM/TMD/455280

Note No:9

Reviewed

Submitted by:

कृष्णा बीसेट्टी - उप महाप्रबंधक

KRISHNA BHESETTY - 009044 (DY. GENERAL MANAGER ,ELECT MAINT- MAIN PLANT)
(BKRISHNA@NTPC.CO.IN, 9493546044) ,Simhadri

On:20/09/2023(10:34:43) **Ref:**SMTTPP/MECH MAINT-/2023-24/SIM/TMD/455280

Note No:10

As discussed.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoor - 005637 (GENERAL MANAGER ,MAINTENANCE)
(MEKOVOOR@NTPC.CO.IN, 9650994652) ,Simhadri

On:09/10/2023(18:03:03) **Ref:**SMTTPP/MECH MAINT-/2023-24/SIM/TMD/455280

Note No:11

Corrections in document made as discussed. Updated LMI uploaded.

May please process further.

Submitted by:

रश्मि रंजन सतपथी - वरिष्ठ प्रबंधक

RASHMI RANJAN SATAPATHY - 100837 (SENIOR MANAGER ,MECH MAINT- TURBINE)
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On:20/10/2023(12:14:55) **Ref:**SMTTPP/MECH MAINT-/2023-24/SIM/TMD/455280

Note No:12

Approval may please be accorded.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoor - 005637 (GENERAL MANAGER ,MAINTENANCE)
(MEKOVOOR@NTPC.CO.IN, 9650994652) ,Simhadri

On:27/10/2023(18:57:42) **Ref:**SMTTPP/MECH MAINT-/2023-24/SIM/TMD/455280

Note No:13

Approved & Submitted by:

संजय कुमार सिन्हा - महाप्रबंधक

Sanjay Kumar Sinha - 005185 (GENERAL MANAGER ,HEAD OF PROJECT)
(SKSINHA06@NTPC.CO.IN, 9431215239) ,Simhadri

On:04/11/2023(18:43:20) **Ref:**SMTTPP/MECH MAINT-/2023-24/SIM/TMD/455280

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/054/

ISSUE NO.1; REVISION NO. 00; DATE: OCT'2023

**INSPECTION AND MAINTENANCE OF
CONDENSER ON LOAD TUBE CLEANING
SYSTEM (COLTCS)**

APPROVED FOR

***Implementation*-----**

HOP, SIMHADRI

Date-----

LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	INSPECTION AND MAINTENANCE OF CONDENSER ON LOAD TUBE CLEANING SYSTEM (COLTCS)
<u>2.a) DOCUMENT CODE</u>	<u>LMI/OGN/OPS/MECH/054/</u>
b) REVISION/ISSUE NO.&DATE	<u>ISSUE-2; REV.-00; OCT'2023</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>OGN/OPS/MECH/054</u>
<u>4.PRERPREARED BY</u>	NAME: R R SATAPATHY SR MANAGER (TMD) SIGNATURE-----
<u>5. CHECKED BY</u>	NAME: S K SUMAN AGM (TMD) SIGNATURE-----
<u>6.RECHECKED BY</u>	NAME: MATHEW EIPE KOVOOR GM(MAINT) SIGNATURE-----
<u>7. REVIEWED BY</u>	NAME: D P PATRA GM(O&M) SIGNATURE-----
<u>8. APPROVED BY</u>	NAME: S K SINHA GGM SIGNATURE-----
<u>9.INDEX LIST</u>	ENCLOSED.
<u>10.DISTRIBUTION LIST</u>	ENCLOSED.

NTPC	LMI	SIMHADRI
TITLE: INSPECTION AND MAINTENANCE OF CONDENSER ON LOAD TUBE CLEANING SYSTEM (COLTCS)		
REV:00 DATE: OCT'2023		
REF NO. LMI/OGN/OPS/MECH/054/		PAGE: 1 OF 13
<u>INDEX</u>		
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2.0	INTRODUCTION	2
3.0	SCOPE	2
4.0	REFERENCE DOCUMENT	2
5.0	SUPERSEDED DOCUMENT	3
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8.0	EXCEPTION REPORTING	5
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NTPC	LMI	SIMHADRI
TITLE: INSPECTION AND MAINTENANCE OF CONDENSER ON LOAD TUBE CLEANING SYSTEM (COLTCS) REV:00 DATE: OCT'2023		
REF NO. LMI/OGN/OPS/MECH/054/		PAGE: 2 OF 13
<u>2.0 INTRODUCTION</u> Condensers play a key role in the process of Power Generation. A lower back pressure of the turbine results in a higher generation of power and this largely depends on the performance of the condensers. It is essential to maintain the heat transfer rate and co-efficient close to the design parameters for improved steam condensation. Hence the cleanliness of the condenser tubes should be kept as close as possible to the design values in order to increase the condenser efficiency which will result in an optimum Power Generation. Condenser on load tube cleaning system (COLTCS) is an effective way to maintain the cleanliness of condenser tubes. Sponge rubber balls with a special composition and approx. size 1 to 3mm greater than the inner diameter of the Condenser Tubes is injected into the Cooling Water stream at the inlet. These balls with specific gravity close to the cooling water are carried into the Condenser along with flow of the inlet water. Cleaning balls are then pushed through the tubes by the differential pressure existing across the Condenser tubes. While passing through the Condenser tubes, the balls are pressed against the walls of the tubes removing deposits on the inner surface of the tubes. A specially designed screen arrangement called the Ball Separator, separates the balls from the main Cooling Water system and the balls are collected in the Ball Vessel. The balls are injected into the Cooling Water through the Ball Recirculating Pump. The main advantages of a Tube Cleaning System are: • Increase in the Heat Transfer Rate • Optimum Turbine Back Pressure • Increased Generation Efficiency • Prevention of corrosion/scaling in condenser Tubes • Need for Shutdown for Manual Cleaning of condenser tubes • Reduction in Fuel costs The objective of this LMI is to prevent scaling of condenser tubes and thus increase the condenser efficiency. <u>3.0 SCOPE</u> The scope of this LMI is to specify general precautions to be taken to maintain the healthiness of COLTC system. <u>4.0 REFERENCE DOCUMENT</u> Operation guidance note OGN/OPS/MECH/054 circulated by operation services-NTPC corporate center.		

NTPC	LMI	SIMHADRI	
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REF NO. LMI/OGN/OPS/MECH/054/		PAGE : 3 OF 13	
<u>5.0 SUPERSEDED DOCUMENT: NIL</u>			
<u>6.0 SPECIFIC INSTRUCTIONS</u>			
To achieve the objective of this LMI the following approach is to be followed:			
a) Periodic Inspection & Maintenance of COLTC system as per the schedule			
<u>7.0 METHODOLOGY OF IMPLEMENTATION AND RESPONSIBILITY CENTERS</u>			
7.1 COLTC system consists of Ball separator, DP measuring system, Ball recirculating skid, Injection & ball transport piping, Cleaning balls, Ball recirculating monitor and Ball monitoring system. In Simhadri, as cooling medium is sea water, titanium tubes are used in condenser. RB 20 or RG 15 rubber balls to be used for titanium tubes as per OEM M/s GEA-BGR.			
7.2 CHECKLIST FOR INSPECTION & MAINTENANCE			
S.No	Operations	Frequency of check	Responsibility
A. Ball separator			
1	Inspect general condition of the screens, screen shaft and funnel assembly	Every Overhaul/ Opportunity	TMD & Operation
2	Check the sealing of the screen bearings.	Every Overhaul/ Opportunity	TMD
3	Check smooth movement of the shafts	Every Overhaul/ Opportunity	TMD & Operation
4	Check whether the indication on the screen shaft is in line with marking on the shaft disc and worm gear labelling at "Operating Position" Or “Backwash Position”	Every Overhaul/ Opportunity	TMD & Operation
5	Carry out limit setting of actuators used for rotation of screen	Every Overhaul/ Opportunity	TMD, EMD, C & I & Operation
6	Check cleaning & thoroughness of pipelines	Every Overhaul/ Opportunity	TMD & Operation
7	Checking of gap between screen & shell in operating position	Every Overhaul/ Opportunity	TMD
8	Check wear and tear of gaskets, bolts & nuts	Every Overhaul/ Opportunity	TMD

NTPC		LMI		SIMHADRI	
TITLE: INSPECTION AND MAINTENANCE OF CONDENSER ON LOAD TUBE CLEANING SYSTEM (COLTCS) REV:00 DATE: OCT’2023					
REF NO. LMI/OGN/OPS/MECH/054/			PAGE: 4 OF 13		
S.No	Operations	Frequency of check	Responsibility		
9	Inspection & cleaning of Extraction arm (ball collecting area).	Every Overhaul/ Opportunity	TMD & Operation		
B. DP Measuring system					
1	Flushing of all impulse lines to DP transmitters	Quarterly	C & I		
2	Calibration of all DP Transmitters	Every overhaul	C & I		
3	Auto opening of screen logic may be converted to alarm and screen should be opened after ball collection cycle	Regular	C&I/Operation		
C. BALL RECIRCULATING SKID					
1	Inspection of shaft, impeller	Every overhaul	TMD		
2	Inspection/Replacement of bearings	Every overhaul	TMD		
3	Inspection of Gland packing/Mech seal	Weekly	TMD		
4	Inspection of oil level	Weekly	TMD		
5	Alignment of pump & Motor	Half yearly	TMD		
6	Coupling Spider replacement	Half yearly	TMD		
D. Injection and Ball transport piping					
1	Ensure cleaning & thoroughness of pipes	Every Overhaul/ Opportunity	TMD & Operation		
2	NDT check of weld joints	Every Overhaul	TMD		
E. Cleaning balls					
1	Ensure proper type of balls are used	Before ball charging	TMD		
2	Ensure ball diameter is more than tube internal diameter	During inspection/ Before ball charging	TMD		
3	Ensure balls are squeezed twice/thrice in 2-3 hours before charging	Regular	Operation		
4	Measurement of ball size with help of external ball sorter sieve	Weekly	Operation		
5	Removal of damaged balls	Every time before charging the ball	Operation		
6	Procurement of cleaning balls with provision of part delivery	Regular	TMD		
7	Ensure storage of cleaning balls in cool area	Regular	TMD		

NTPC		LMI		SIMHADRI	
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REV:00 DATE: OCT’2023					
REF NO. LMI/OGN/OPS/MECH/054/			PAGE: 5 OF 13		
S.No	Operations	Frequency of check	Responsibility		
F. BRM & BMS					
1	Ensure BRM & BMS are in service	Regular	C & I and Operation		
G. Condenser inlet & outlet butterfly valves					
1	Replacement of door seal of all inlet and outlet valve followed by joint inspection to ensure isolation for online water box cleaning	Every overhaul	TMD & Operation		
2	Limit setting checking of all CW inlet and outlet valves to ensure proper closing	Every Overhaul	C&I, EMD, Operation & TMD		
H. Condenser water box cleaning					
1	Condenser water box cleaning after isolation of one pass	Opportunity	Operation & TMD		
2	Condenser water box cleaning during unit outage of more than 02 days	Opportunity	Operation & TMD		
I. Condenser tube cleaning					
1	Condenser tube jet/Bullet cleaning in available opportunity as per COS circular	Opportunity	TMD		
<u>8. EXCEPTION REPORTING</u>					
Any exceptions to the instructions specified in the LMI are to be reported to HOD (MM) as per the following format					
Responsibility- HOD (TMD)					
Unit No.	Description of exception	Details	Remarks		
<u>9.0 RESPONSIBILITIES</u>					
The responsibility of carrying out different tasks as detailed in the LMI are with HOD (TMD), HOD (Operation) and HOD (C&I) & HOD(EMD).					
<u>10. RECORD KEEPING</u>					
The records to be maintained under this LMI include					
a) Protocols jointly signed during overhauls b) Exception reporting, if any					
The above records are to be maintained by MM- TMD					
<u>11.REVIEW</u>					
This document shall be reviewed by HOP, SIMHADRI on two-yearly basis.					

[illegible]

NTPC	COMPLIANCE RECORD (PROTOCOL-1)		SIMHADRI																																										
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NTPC	COMPLIANCE RECORD (PROTOCOL-3)			SIMHADRI																									
TITLE: INSPECTION AND MAINTENANCE OF CONDENSER ON LOAD TUBE CLEANING SYSTEM (COLTCS)																													
REV:00 DATE: OCT'2023																													
REF NO. LMI/OGN/OPS/MECH/054/			PAGE: 10 OF 13																										
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NTPC		COMPLIANCE RECORD (PROTOCOL-4)		SIMHADRI	
TITLE: INSPECTION AND MAINTENANCE OF CONDENSER ON LOAD TUBE CLEANING SYSTEM (COLTCS)					
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S.No.	Description			Status	
A. DP Measuring system					
1	Auto opening of screen logic may be converted to alarm and screen should be opened after ball collection cycle				
B. BRM & BMS					
1	Ensure BRM & BMS are in service				

NTPC		COMPLIANCE RECORD (PROTOCOL-5)		SIMHADRI													
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NTPC	COMPLIANCE RECORD (PROTOCOL-6)	SIMHADRI															
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